NeuroCore – Core Research Summary

Objective

Define and validate a transition-sensitive, domain-agnostic framework for identifying functional regime changes in time-series data, with initial validation on neurophysiological datasets (EEG).

Core Concept

The system models a signal $x(t)$ as evolving within a functional space X, partitioned into regimes Γ_k. A transition-sensitive operator $\Phi(t)$ extracts local dynamics, enabling inference of regime states $\lambda(t)$.

Key Operators

1. $\Phi(t) = \Sigma \log(1 + |\nabla x(i)|) \rightarrow$ local dynamics integration
2. $\Delta(t) = |\Phi(t) - \Phi(t-1)| \rightarrow$ transition sensitivity
3. $K(t) = \sigma / \mu \rightarrow$ structural coherence / order parameter
4. $L(x(t)) \rightarrow \phi \approx 0.55 \rightarrow$ asymptotic stability invariant

Primary Datasets (Validated)

1. CHB-MIT (Epilepsy EEG)
2. EEG Motor Movement/Imagery (AUC ≈ 0.83)
3. Siena Scalp EEG

Key Results

1. AUC up to ~0.83 on EEG classification tasks
2. Transition-sensitive features outperform simple baselines
3. Cross-dataset applicability demonstrated
4. Evidence of pre-transition signal structure (to be validated rigorously)

Validation Strategy (Next Phase)

1. Nested cross-validation
2. Permutation testing
3. Baseline comparison (variance, entropy, etc.)
4. Event-aligned transition significance testing

Core Files to Keep

1. neurocore_chbmit_functional_stability_v1.ipynb
2. siena_pipeline_results.ipynb
3. ASHI-CORE_TEIZEL.ipynb

4 DOC_PAPER.ipynb

Files to Remove / Archive

Remove unrelated datasets, experimental notebooks without results, and non-neurophysiological data to reduce noise and improve clarity.

Conclusion

The NeuroCore framework shows promising capability in detecting functional transitions in time-series data. To reach publication-level maturity, focus must shift toward rigorous validation, reproducibility, and clear benchmark comparisons.